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Review and synthesis

Multi-species occupancy models: review, roadmap, and recommendations

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Recent technological and methodological advances have revolutionized wildlife monitoring. Although most biodiversity monitoring initiatives are geared towards focal species of conservation concern, researchers are increasingly studying entire communities, specifically the spatiotemporal drivers of community size and structure and interactions among species. This has resulted in the emergence of multi-species occupancy models (MSOMs) as a promising and efficient approach for the study of community ecology. Given the potential of MSOMs for conservation and management action, it is critical to know whether study design and model assumptions are consistent with inference objectives. This is especially true for studies that are designed for a focal species but can give insights about a community. Here, we review the recent literature on MSOMs, identify areas of improvement in the multi-species study workflow, and provide a reference model for best practices for focal species and community monitoring study design. We reviewed 92 studies published between 2009 and early 2018, spanning 27 countries and a variety of taxa. There is a consistent under-reporting of details that are central to determining the adequacy of designs for generating data that can be used to make inferences about community-level patterns of occupancy, including the spatial and temporal extent, types of detectors used, covariates considered, and choice of field methods and statistical tools. This reporting bias could consequently result in skewed estimates, affecting conservation actions and management plans. On the other hand, comprehensive reporting is likely to help researchers working on MSOMs assess the robustness of inferences, in addition to making strides in terms of reproducibility and reusability of data. We use our literature review to inform a roadmap with best practices for MSOM studies, from simulations to design considerations and reporting, for the collection of new data as well as those involving existing datasets.

Keywords: biotic interactions, community models, hierarchical models, multi-species models, occupancy, reporting bias

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Introduction

Monitoring the distribution and abundance of wildlife populations is an essential part of advancing ecological understanding and effective conservation management. However, the majority of monitoring initiatives have been geared towards a single focal species, typically those of conservation concern or management interest, likely due to the resource-intensive nature of wildlife monitoring and a lag in statistical and computational tools and methods to handle ecological data. With the growing concern over the effects of global change on community composition, information at the community level is increasingly necessary. At the same time, advances in statistics and the increasing availability of data across taxa are enabling analyses at the community scale (Fig. 1).

The burgeoning interest in making inferences about communities has led to analytical approaches that utilize data on all observed species, including bycatch species that are obtained incidentally during the course of a study that was designed with one or more focal species in mind (Iknayan et al. 2014, Guillera-Arroita 2017, Ovaskainen et al. 2017, Steenweg et al. 2019). While the design is optimized for the focal species, there is often information about other species readily or potentially available for extrapolation. Recently, there has been rapid development of statistical methodologies to investigate important aspects of community ecology (Iknayan et al. 2014, Jarzyna and Jetz 2016, Ovaskainen et al. 2017). For example, some community ecology projects now formally incorporate imperfect detection and spatial variation in detectability and occurrence patterns (Dorazio and Royle 2005, Zipkin et al. 2010). These approaches have also been applied to deriving biodiversity metrics and studying changes in community size, composition and function in space and time (Dorazio et al. 2006, Tobler et al. 2015, Jarzyna and Jetz 2016) as well as interactions among species (Waddle et al. 2010, Pollock et al. 2014, Rota et al. 2016, Tobler et al. 2019).

Currently, the majority of community models, especially hierarchical models that infer state variables of undetected species while accounting for imperfect detection, have great flexibility in incorporating variation in detectability and occurrence. However, they do not formally account for concerns that result from using a single design to study heterogeneous communities. Recent reviews have attempted to evaluate certain assumptions of community models (Linden et al. 2017, Guillera-Arroita et al. 2019); however, the assumption that the designs are adequate for each species in community analyses, an inherent feature of community monitoring initiatives, has not yet been investigated.

We reviewed the literature on one of the most widely applied classes of community models, the hierarchical multispecies occupancy model (MSOM) (e.g. Dorazio and Royle 2005 and extensions). We contrast existing approaches to MSOMs and critically evaluate the assumptions of MSOMs in order to identify the potential for biases when studying communities that vary in terms of their spatial ecology and

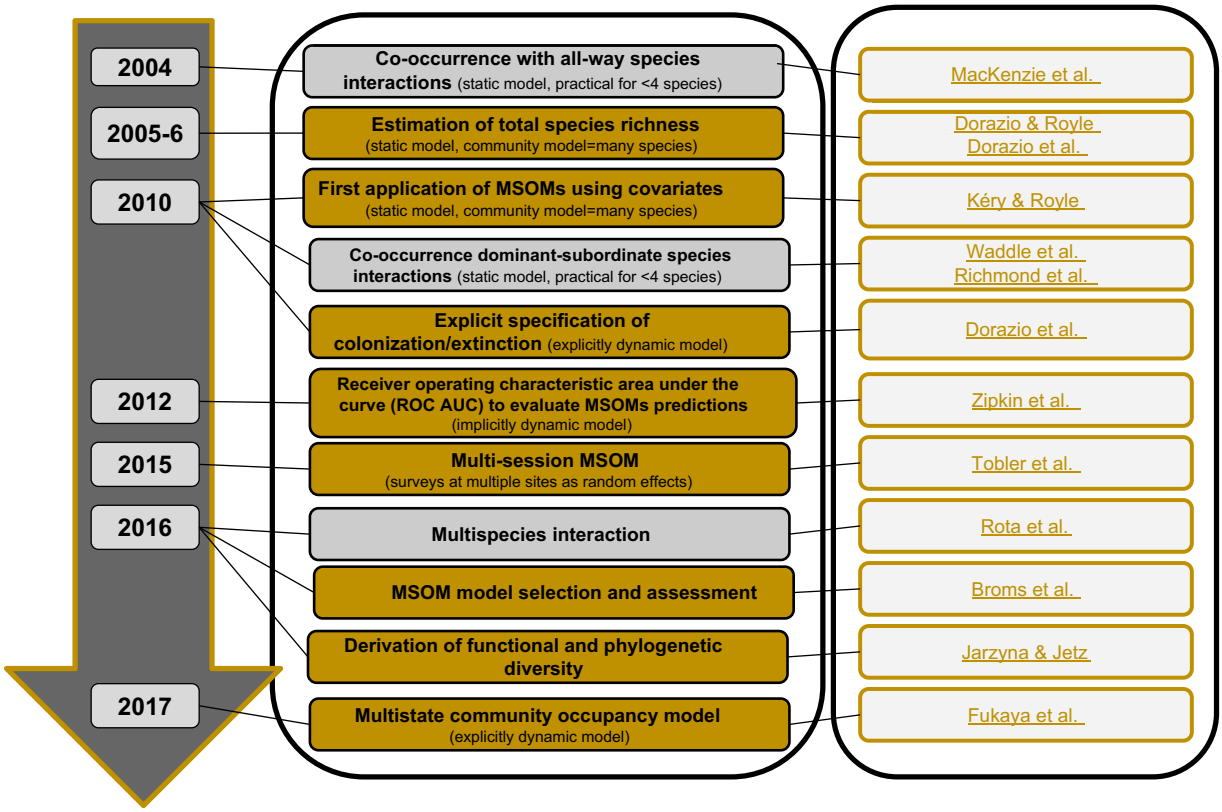


Figure 1. Key methodological developments in the history of MSOMs. The developments related to species interactions are in grey boxes.

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species traits. We use this theoretical approach to develop a set of criteria to consider in the design phase of a community monitoring scheme. Next, we develop a set of recommendations to aid in the design of future community monitoring efforts. We outline a workflow of best practices from definition of the study design to inference and (meta)data reporting that can be adopted as a reference model in multi-species studies. Finally, we report additional observations from the review to highlight aspects of community studies that will facilitate better application and reporting practices in the future.

Multi-species occupancy models: state of the field

There are different types of MSOMs, based on study objectives. Often, community occupancy and multi-species occupancy are used interchangeably, although subtle differences exist (Guillera-Arroita 2017). MSOMs encompass both co-occurrence models that incorporate interactions between a few species (typically coupled dominant–subordinate interactions) as well as community models that usually involve some estimate of species richness without including biotic interactions (Fig. 1). Co-occurrence models include biotic interactions in addition to habitat and abiotic factors, and modeling the detection and occurrence hinges on species traits and the nature of the interactions. Frequently, assumptions on the nature of the interactions need to be made in order to model them appropriately. These could take the form of competitive interactions or prey–predator systems, and assumptions made include body size-based hierarchy in the event of a dominant–subordinate species coupling (Fig. 1).

The MSOM can jointly estimate detection and occurrence probabilities of each species in the community, including those not detected. As with the single species occupancy model (SSOM), MSOMs have at their core the fundamental inference objective of estimating species occupancy rates in the presence of imperfect detection (Box 1). This requires repeated sampling of multiple spatial locations (MacKenzie et al. 2002, 2017).

There are many limitations to MSOMs and several are related to the statistical framework used. Bayesian inference can be challenging in terms of being computationally intensive, with model selection being problematic (Tenan et al. 2014, Broms et al. 2016). These types of models often entail considerable training to code and interpret. Furthermore, incorporating interactions has consistently been a challenge, requiring careful deliberation about the variables used.

Model assumptions under community sampling

Data collected on multiple species by a particular array of detectors (e.g. trail cameras, traps, transects and point counts) are a function of within-community variation in species-specific traits such as mobility, density, habitat requirements and behavior, as well as inter- and intra-specific interactions. For reliable inferences, the resulting data must meet, to varying levels of strictness, specific model assumptions (MacKenzie et al. 2017): demographic and geographic closure, independence, correctness in identification and ecological similarity. Whether these assumptions are met is related to the ecological traits of the species of interest and the spatiotemporal characteristics of the monitoring design. It follows, therefore, that inherent variability in ecological traits within a community produces variability in the extent to which assumptions are violated or adhered to. For instance, the home range size of the species is correlated tightly with the assumptions of geographic closure and independence. Therefore, the issue of whether community-level inferences are reliable is more of a design-based than a model-based consideration, yet current developments and publication trends would suggest the reverse.

The basic assumptions of an MSOM are similar to those of the SSOM and include:

Geographic and demographic closure

The sites represent closed populations, and there are no births, deaths, colonization or extinction in the span of the surveys. When this assumption is violated, there tends to be an overestimation of occupancy while detection is low, which leads to inflated species richness estimates (Kéry and Royle 2015). This assumption can be maintained with the help of a priori simulations that inform study design and assist in identifying the spatial and temporal replicates necessary, as well as being cognizant of detector attributes with regard to species traits such as home ranges, rarity, elusiveness and mobility (Linden et al. 2017).

Independence

At a site, the probability of detecting a species and that of the site being occupied are independent of the occupancy and detection probability of another site. Violations of this assumption typically happen when there is oversampling or clustered sampling (instead of a random sampling scheme), especially while using camera traps and acoustic recording equipment, during studies involving rare and elusive species. The number of sampling units as well as the sampling methodology have an impact on whether this assumption is violated, while spatial correlations often do not result in a violation. Any autocorrelation and violations of this assumption can be tested statistically (Wright et al. 2016). A violation of this assumption can lead to overestimated precision for species occupancy, detection and richness (McNew and Handel 2015). One solution proposed to address this problem and decrease uncertainty around the estimates involves modifying the detection-level component of the model (Wright et al. 2016). This could, for instance, involve determining detector spacing during study design based on the home ranges of the study

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Box 1. Anatomy of a multi-species occupancy model

Hierarchical occupancy models are used to estimate the number of sites that are occupied by species of interest in a landscape in an attempt to understand species distribution patterns. The detectability of the study species forms an essential component of occupancy studies, especially in trying to incorporate variation in the detection of the species at different sites. These models involve a separation between the state (occupancy) and observation (detection) processes (MacKenzie et al. 2002).

They can be analyzed using either Bayesian or frequentist approaches, with the former gaining popularity in recent years. Integral to either approach is the detection probability (p) and the occupancy probability (ψ). Parameter p is the probability of detecting the species conditional on the presence of the species at that particular site, and ψ is the probability that a site is occupied by a species. Spatial ('sites') and temporal ('visits') replication is also integral to occupancy modeling.

The study area is a mosaic of observed occupied and unoccupied sites. However it is difficult to determine whether a site where a species was not detected is truly unoccupied. It is possible the species was not detected due to various reasons such as insufficient visits (temporal replicates) and variation in species detectability. This variation in detectability, which is affected by species traits and survey characteristics, is known as imperfect detection.

Imperfect detection of species could be due to a variety of factors that are biotic or abiotic. These are typically related to species traits (life history, behavior, population density, sex, age, trap shyness), habitat characteristics (vegetation type, terrain, elevation, noise, canopy cover) or survey-related factors (sampling effort, observer experience, method used, spatial and temporal scales, time of the year). Due to their effectiveness in accounting for imperfect detection, occupancy models have become a mainstay of species distribution studies since they were first introduced. While they have been used globally on a range of taxa using a variety of field methods, studies in the first decade of their introduction typically focused on a single, focal species.

Like the classical single species occupancy models (SSOMs), MSOMs are built on the encounter histories of species across sites in a region during repeated visits. They are typically used to estimate species richness (N) at both the community and metacommunity level. MSOMs are based on the concept that a community is an ensemble of species occurring at a site and a metacommunity is the collection of such communities.

MSOMs combine three levels of hierarchy: metacommunity, community, and individual species. The first level describes the presence of each species $i = 1, 2, \ldots, N$ in the metacommunity through an indicator variable (w_i):

$$w_i \sim \text{Bernoulli}(\Omega) \tag{1}$$

where Ω is the probability that species i is a member of the metacommunity of size N.

The parameter-expanded data augmentation (DA) technique is typically used to implement the model in a Bayesian framework with Markov Chain Monte Carlo (MCMC) methods (Dorazio and Royle 2005). The DA technique involves the addition of an arbitrary $M \gg N$ number of all-zero trap frequencies, which can be seen as potentially unobserved species, to the detection matrix Y. In this way, the number of unobserved species in each community can be estimated by evaluating which of the $M - N$ species (rows of the augmented matrix) are members of the metacommunity (sampling zeros, $w_i = 1$) or not (structural zeros, $w_i = 0$). Data augmentation converts the problem of estimating N into the equivalent problem of estimating Ω and species richness N is computed as a derived parameter by summing up the latent indicators w_i since the expectation of N is equal to $M\,\Omega$ (Kéry and Schaub 2012).

The second level of hierarchy of an MSOM defines the occurrence of each species at specific sites $j = 1, 2, \ldots, J$, i.e. at the community level (ecological or state process):

$$z_{ij} \sim \text{Bernoulli}(\psi_{ij} w_i) \tag{2}$$

where z_{ij} are elements of the true occupancy state matrix Z, of dimension M by J, indicating whether species i is present at site j ($z_{ij} = 1$) or not ($z_{ij} = 0$). The parameter ψ_{ij} represents the species-specific occupancy probability at each site. Note that matrix Z can be used to derive (meta)community-level metrics.

The third level describes the encounter process of individual species and the related false-negative measurement error at each site sampled $k = 1, 2, \ldots, K$ times:

$$y_{ij} \sim \text{Binomial}\left(K, p_{ij} z_{ij}\right) \text{ or } y_{ijk} \sim \text{Bernoulli}\left(p_{ijk} z_{ij}\right) \tag{3}$$

where y_{ij} indicates the detection frequency of species i at site j over K sampling occasions, and y_{ijk} is the detection/non-detection at the kth sampling occasion, with detection probability $p_{ij(k)}$.

Occupancy and detection probabilities can be modeled as a function of covariates ('cov'):

$$\text{logit}\left(\psi_{ij}\right) = \alpha_{\psi,i} + \beta_{\psi,\text{cov},i} \text{cov}_j \tag{4}$$

(Continued)

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Box 1. Continued

$$\text{logit}\left(p_{ijk}\right) = \alpha_{p,i} + \beta_{p,\text{cov},i}\text{cov}_{jk} \quad (5)$$

Further model assumptions are defined to characterize heterogeneity in occurrence and detectability among species, and to estimate occurrence of species members of the community not observed in any site. Often, species are assumed to have a similar, but not identical, response to environmental changes (Dorazio et al. 2011, Kéry and Royle 2015). In the case of interspecific interactions, where occurrence of a species is affected by the presence or absence of another species, patterns of co-occurrence should be considered in lieu of ecological similarity (MacKenzie et al. 2004). When ecological similarity is assumed, unobserved sources of heterogeneity in occurrence and detection among species can be modeled by adding a further hierarchical level and assuming that species-specific parameters in the related linear predictors (α_i and β_i in Eq. 4, 5) are independent random effects: $\alpha_{\theta,i} \sim \text{Normal}\left(\mu_{\alpha_\theta}, \sigma^2_{\alpha_\theta}\right)$ and $\beta_{\theta,\text{cov},i} \sim \text{Normal}\left(\mu_{\beta_{\theta,\text{cov}}}, \sigma^2_{\beta_{\theta,\text{cov}}}\right)$ with $\theta = (\psi, p)$. In this way, species-specific estimates are 'shrunk' toward the mean parameter value of the community. A correlation structure between occupancy and detection probability can also be included if detectability is expected to increase as species abundance increases (Dorazio 2007).

species, identifying adequate number of replicates and detectors, and the optimal distance between detectors for the community study.

Accurate identification

Errors in species identification, either due to fluctuations in detectability or observer error, can significantly impact and bias occupancy estimates (Dorazio et al. 2011). This assumption could be violated in surveys where the detection and identification of species is through their vocalizations (e.g. amphibians, birds, insects) where misidentifications often occur (Simons et al. 2007, McClintock et al. 2010). This is also problematic in situations where the skill level of the observers is variable (Genet and Sargent 2003). A violation of this assumption could result in an overestimation of species through erroneous detection at sites where they were actually absent. One mechanism to address this issue is through the inclusion of parameters for false-positives as well as false-negatives in detection (Royle and Link 2006), although this may not be effective where the misidentification and detection probabilities are similar.

Ecological similarity

Often community monitoring studies assume that species in the community are similar, resulting in species-specific random effects that are drawn from the same distribution, typically the bivariate normal distribution (Kéry and Royle 2015). A consequence of violating this distributional assumption is a deterioration in the occupancy, detection and richness estimates, resulting in prediction errors. This assumption can be maintained by keeping in view the problem statement and the inferences made.

A review of the community occupancy modeling literature

We conducted a literature review on MSOMs with the objectives of identifying weaknesses in current implementations and helping inform the ideal composition of these models. The Web of Science portal was used to identify literature pertinent to multi-species models. The topic search term used in May 2018 was: 'multi-species occupancy' OR 'multispecies occupancy' OR 'community occupancy' OR 'two-species occupancy' OR 'joint occupancy' OR 'co-occurrence of interacting species'. This resulted in a total of 106 publications, of which 92 were relevant to this review.

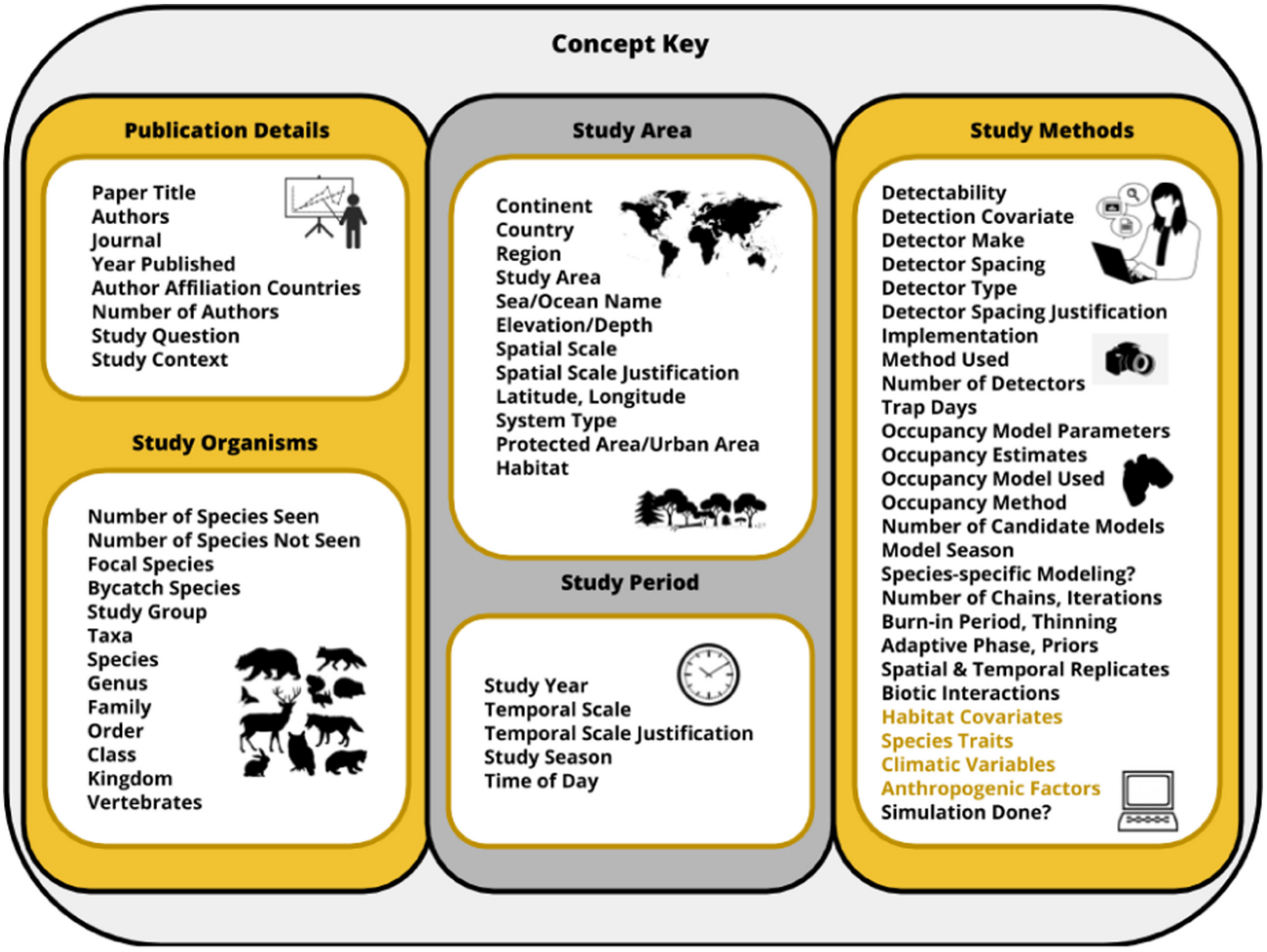
These were then accessed through the multimedia data manager ViXeN (Ramachandran and Devarajan 2018) as a project. Each publication was viewed and the tags annotated. The entered data were then exported as a comma separated value (csv) file for subsequent analysis. The analysis was done using the statistical package R ver. 3.4.4 (2018-03-15; The R Foundation for Statistical Computing) (R Core Team), which was also used for cleaning the review data, analysis and visualization (Wickham and Chang 2008, Wickham 2010, RStudio Team 2018). The tags were broadly classified into study question, study organisms, study area, study period, study methods, covariates and publication details (Fig. 2). The broad ecological and conservation context was inferred based on the study questions.

There were 92 publications involving field-based multi-species occupancy studies published between 2009 and 2018 (Fig. 3) considered for this systematic review. The number of studies for the year 2018 only cover the first half of the year when data collection for the review was completed. Based on the year of publication, there seems to be a general and consistent increasing trend in studies involving MSOMs (Fig. 3).

The field studies spanned six continents (Antarctica is the exception) and 27 countries (Fig. 4). However, a majority of the studies were conducted in North America, particularly the USA (Fig. 4), suggesting a geographic bias in MSOMs. Moreover, marine and aquatic communities were vastly underrepresented suggesting a taxonomic bias towards terrestrial organisms. Vertebrates are much better covered than other taxa. These trends have been highlighted as issues for biodiversity data (Geijzendorffer et al. 2016).

The focus of the studies was variable. Many of the studies involving MSOMs were interested in questions relating

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Variables

Climatic	Habitat	Species Traits	Anthropogenic
Precipitation	Farmland	Forage	Hunting
Hydroperiod	Forest	Nest Type	Logging
Inundation	Grassland	Diet	Agriculture
Solar Radiation	Rainforest	Body Mass	Fragmentation
Fire	Wetland	Home Range Size	Road
Temperature	Savanna	Gestation	Housing Density
Cloudiness	Pasture	Resident	Development
Elevation	Plantation	Migrant	Fire
Flooding	Duneland	Activity Pattern	Plantation
Drought	Shrubland	Dispersal	Recreation
	Marsh	Clutch Size	Deforestation
	Ditch		
	Stream		

Figure 2. Concept key and tags included in the literature review. The variables considered as important predictors by studies used in the review are categorized into climatic, habitat, species traits, and anthropogenic variables in the table below the concept key.

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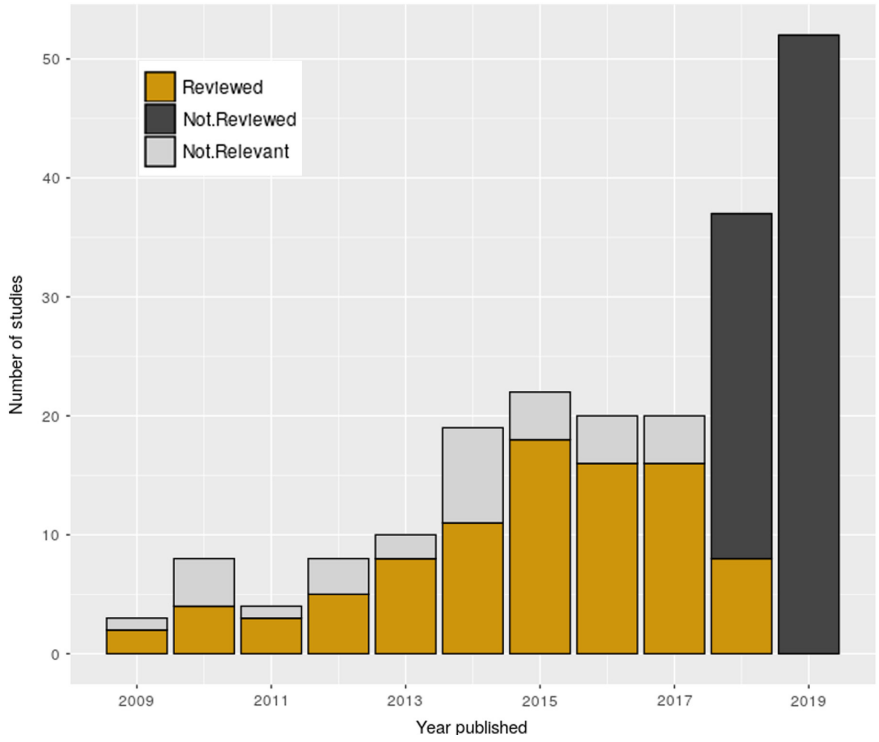


Figure 3. The trend in publications covering multi-species occupancy models over the years. The literature review covers studies that were published until April 2018. The light grey bars represent the number of studies that were not considered for the review, including those that were outside the scope of the review.

to species richness, species conservation and habitat conservation. Total species richness or taxonomic diversity, along with phylogenetic and functional diversity, is considered to be extremely influential for conservation, management and policy decisions (Guillera-Arroita et al. 2019). Although species richness was mentioned to be the focus of about 40% of the studies in the review, most of them (84%) do not report the total number of species found in the study.

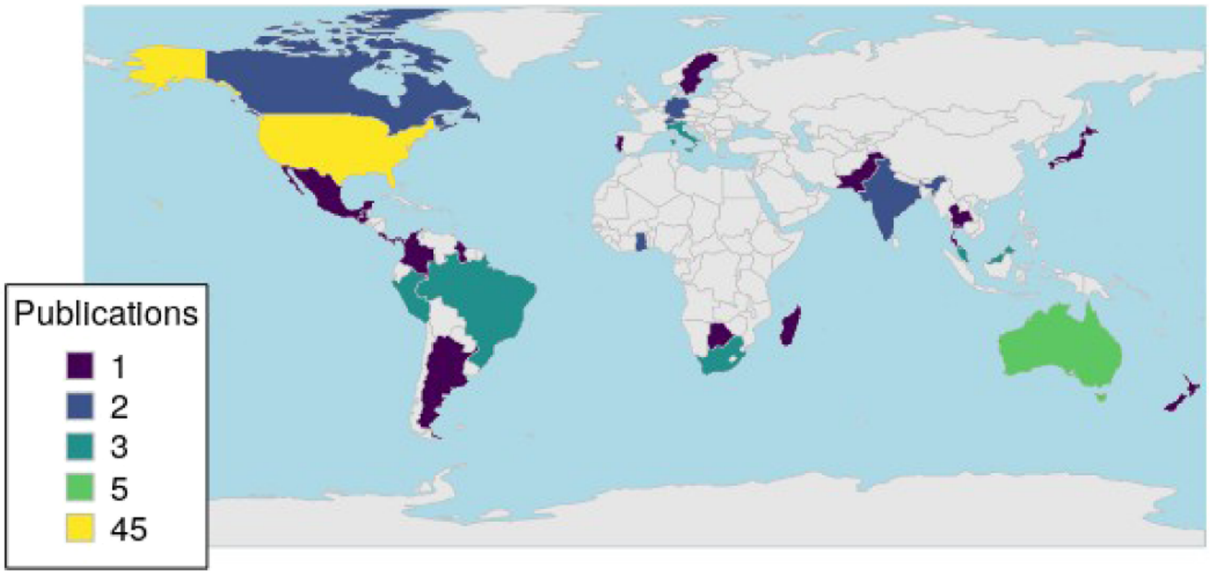
Given the importance of study design in occupancy studies (Guillera-Arroita et al. 2010), the transfer of design considerations is unlikely to be seamless between the focal and bycatch species. It is necessary to account for these disparate design requirements, or at least acknowledge the biases and errors associated with this in MSOMs involving bycatch data. About a quarter of the studies (27 of 92) considered in the review mention the presence of other species in the area which can be considered bycatch, albeit without providing a list outside of the focal species. Approximately a third of the studies (31%) provide an estimate of the number of bycatch species that were found.

However, it is difficult to glean exactly how many of the publications analyzed bycatch species from a study that was originally designed for different focal species, although some of the studies alluded to this (24 out of 92 studies). The actual number of bycatch species ranged from 1 to 472 (Kingdom: Plantae), with a majority of studies mentioning less than twenty.

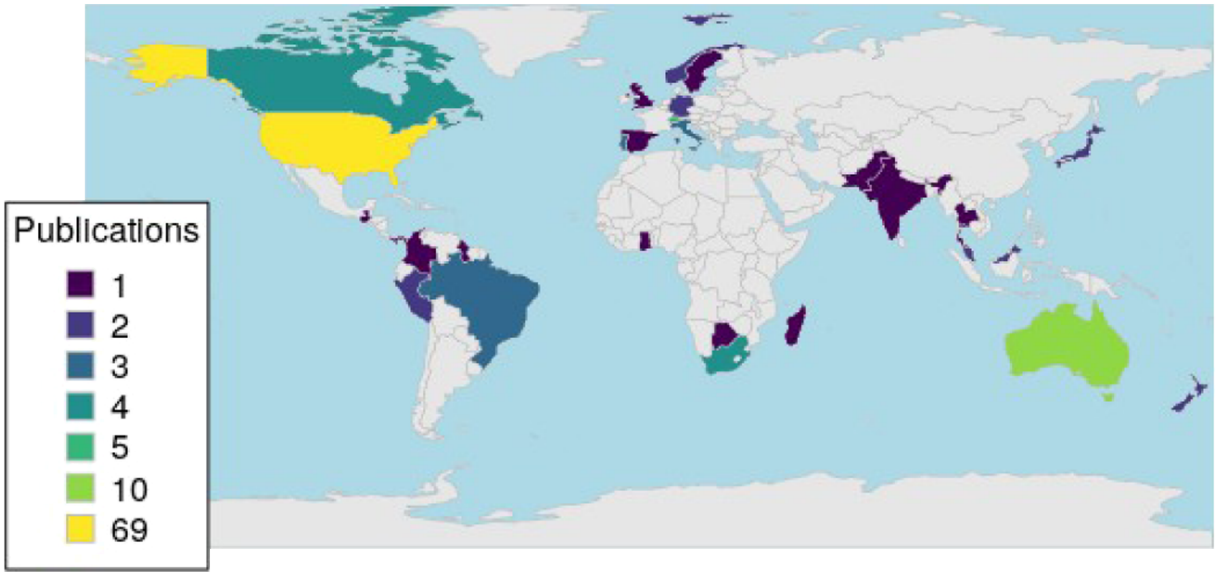
There might be an underestimation of species richness when relevant predictors are not included in the model, such as the habitat covariates used, the choice of detection covariates, and whether biotic interactions need to be considered for occupancy or detection (Guillera-Arroita et al. 2019). Biotic interactions are crucial in community ecology studies and are known to be important predictors of species distribution patterns (Pollock et al. 2014, Rota et al. 2016, Ovaskainen et al. 2017). Species detectability needs to be factored in as well so as to reduce the number of species missed (Dorazio and Royle 2005, Zipkin et al. 2010, Karenyi et al. 2016). Despite the critical nature of detection covariates, as many as 15 papers (16%) did not explicitly mention which detection covariates were used. Often the rationale behind the choice of predictors was not mentioned either. Only 28% of the publications in this review mentioned considering biotic interactions for either occupancy or detection. A majority of the studies in the review did not explicitly state the detectability of the species, not even qualitatively (e.g. the species was easily detected).

Reporting on spatial and temporal aspects of the studies was also lacking. Of the studies considered in the review, almost a quarter (23%) did not report the number of sampling occasions or temporal replicates, 15% made no mention of a temporal scale for the whole study, and close to 90% did not provide any justification on their choice of temporal scale. The same rationale applies to the spatial scale and number of sampling sites or spatial replicates as well. While

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(a) Global distribution of field locations of studies considered in the review



(b) Global distribution of author affiliations of publications considered in the review

Figure 4. Study field locations and author affiliations of multi-species occupancy studies of the publications considered in the review indicate a match between the two. Field studies tended to have at least one collaborator from the country who is an author on the publication, with a few exceptions, such as a mismatch in parts of Central and South America, and disproportionately more authors from North America and Australia included in publications overall.

a majority of the studies (88%) report the number of spatial replicates, approximately half of them (44 out of 92 studies; 47%) have no mention of the spatial scale and only 15% of the total studies provide any justification for the spatial scale of the study. Most of the studies were conducted in protected areas which could be a potential source of bias since inference is restricted to a single type of area and under specific management conditions (namely, protected). In the same vein, most studies were conducted in the summer (Supplementary material Appendix 1), which could also result in inference and reporting biases.

Incomplete reporting was a problem in other areas as well. Only a small fraction of the studies (15 out of 92) even mention the number of species that went undetected during the course of the research. While nearly all the studies provide some information on the detectors used, 57% provide no information on the detector make and 66% do not state the spacing between detectors. Nearly a quarter (23%) of the

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studies provide little or no information on the implementation of the study and sampling methodology employed, such as whether they followed a systematic, random or stratified random sampling approach. Decisions related to the detectors used including the type of detector, spacing between detectors and the implementation (sampling methodology) need to be given considerable thought. Detector spacing is typically calibrated in relation to a specific focal species and thereby affects the detection of bycatch species with different space usage.

Typically, the longer the monitoring, the higher the chance of detecting most or even all of the species present in an area (Guillera-Arroita et al. 2019), as long as the assumptions of demographic and geographic closure are met. In some cases, more visits or temporal replicates do not necessarily correspond with higher detection, and there is a trade-off between effort and detection estimates. By and large, however, the temporal scale is positively correlated with a reduction in the possibility of missing species (Guillera-Arroita et al. 2019). Given this, a justification of the temporal scale chosen and employed in MSOMs is of immense importance.

It was challenging to identify the studies that evaluated assumptions of demographic and geographic closure. These are crucial assumptions of MSOMs, given the inevitable differences in life-history traits among species in a community, on which reporting was extremely sparse. An explicit method to test closure is by fitting an open MSOM using Pollock's robust design framework (Pollock 1982) over short time intervals and evaluating the parameter constraints (Rota et al. 2009). An indirect way to glean this information from the studies is by checking if studies conducted Goodness-of-Fit (GOF) (Broms et al. 2016) tests or temporally thinned their data to see if signs changed. However, this was challenging as well since the use of GOF tests and temporal thinning of data went largely unreported (Zipkin et al. 2009). Alternatively, staggered-entry SSOMs (Kendall et al. 2013) which permit inference about the within-season patterns of species arrival and departure at sampling sites (Chambert et al. 2015) could be used to test closure of the system.

The review shows a slant in the literature towards the use of Bayesian models for obtaining multi-species occupancy estimates, with about 65% of the studies implemented using some instance of BUGS, JAGS or Stan. The choice of priors is considered crucial in Bayesian analysis and has been found to bias estimates (Guillera-Arroita et al. 2019). For instance, choosing incorrectly informative priors is likely to result in a frequentist bias. Despite the importance of prior definition in the Bayesian framework, most studies do not elaborate on the decision process involved. Only half of the studies using Bayesian methods reported the kind of prior that was used, with more than 70% of these (24 out of 33 studies) indicating uninformative priors. Lastly, simulation studies are considered important in evaluating the sensitivity of the estimates obtained in the application to empirical data (Kleijnen 1998, Peck 2004). Only 14 of the 92 publications considered for this review indicate having used simulations in order to gauge model sensitivity.

Best practices for MSOMs

Based on our literature review, we propose a framework outlining best practices for implementing an MSOM (Fig. 5). The MSOM should start with a simulation study which is done a priori based on the researchers' knowledge of the study system, in order to calibrate the study design. MSOMs will be more or less sensitive to some factors such as detectability, detector spacing and certain covariates. The study needs to be carefully designed as not to violate assumptions of closure; considering timing of reproduction and dispersal is key.

The study question determines the design. During this phase, it is important to ascertain whether the study question can rely on existing data (e.g. citizen science and bycatch datasets) or whether new data need to be collected. Appropriate detection methods would have to be chosen based on the nature of the study. This would, in turn, lead to deciding on adequate spacing between the detectors, as well as the spatial and temporal replicates necessary. It is also important to be cognizant of heterogeneity in observers and detectors, which can subsequently help account for imperfect detection.

If the ecological system involves focal species, species characteristics such as home ranges and space usage, detectability, and behavior will need to be considered. On the other hand, if the focus is a community of species or in situations where by-catch species from an existing study are considered, community characteristics will need to be taken into account. These include taxonomic, phylogenetic and functional diversity, as well as species detectability and traits distribution, and interactions among species (Pollock et al. 2017, Guillera-Arroita et al. 2019). In addition to the species and community characteristics, it is crucial to consider site characteristics as well as the survey conditions which account for any spatio-temporal variation in the system (Guillera-Arroita 2017). Taken together, these would comprise the variables thought to influence occupancy and abundance.

The measurements collected for these covariates would form the data which feed into the model. The community data collected should ideally have some estimate of the proportion of missed species as well as the bycatch data used. This is necessary since not accounting for these could result in poor species richness estimates, leading to inference bias (Guillera-Arroita et al. 2019). Similarly, during the modeling process, adequate thought needs to be given to the choice of priors and parameter assumptions. Failure to do so could result in overestimation of species richness estimates. If there were any missing covariates, or in situations with low detectability or abundance, there could be inference bias in the form of underestimation of species richness. Lastly, not accounting for biotic interactions could result in biased richness, occupancy and abundance estimates (Tobler et al. 2019).

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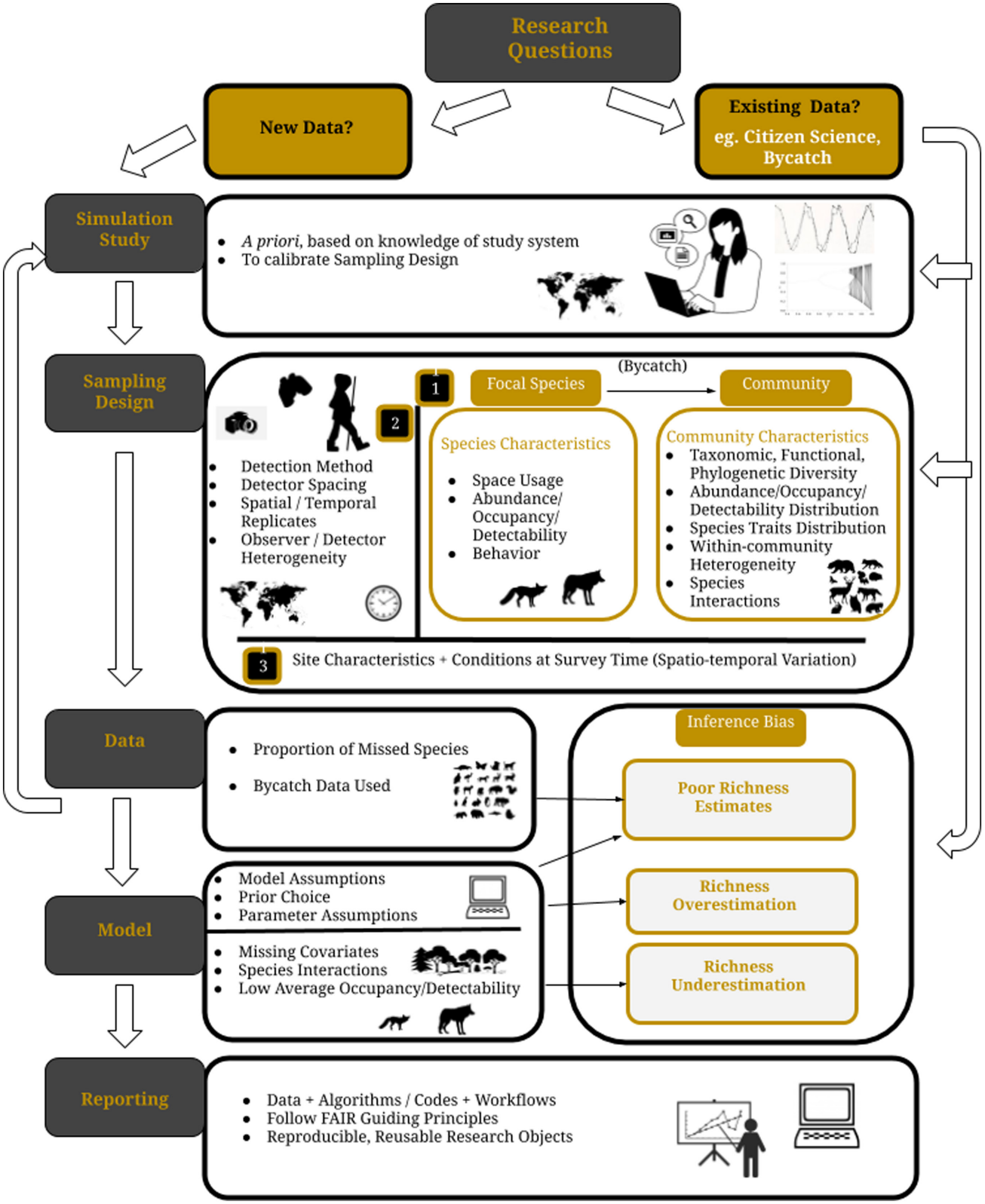


Figure 5. A schematic workflow and roadmap for best practices while implementing an MSOM. Conceptual representation of an ideal protocol for multi-species studies, from a priori simulations to reporting results. The research questions and objectives for the study would determine the type of data that are likely to be used for obtaining the occupancy estimates; in case of existing data, such as from citizen science projects or involving bycatch from a study designed for focal species, the steps followed would differ from when a study is planned for focal species from the ground up.

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Finally, once the study has been completed, when publishing the research, the data, algorithms, and workflows used should be reported (Powers and Hampton 2019). We recommend following the FAIR (findability, accessibility, interoperability and reusability) guiding principles for managing scientific data (Wilkinson et al. 2016). Scientific data may be standalone, citable products due to which the research objects such as the methodology, algorithms, and workflows used should be reproducible, and the data and code reusable. This makes the data easier to find and access, and enhances the authenticity of the research. Issues such as the lack of information on the statistical modeling aspects such as the priors used can be minimized by sharing the supplemental code for running the model. Similarly, information on whether model validation was done and checking for any violations of model assumptions ought to be provided as supplemental code.

Conclusions and synthesis

MSOMs are increasingly being used to monitor species distribution and abundance (Beaudrot et al. 2016). The assumptions and limitations of the traditional single species models propagate to the multi-species framework, with some additional factors that need to be considered. MSOMs are important since they are resource efficient, capable of accounting for biological interactions such as competition and predation, used for existing data, and can be used to monitor entire landscapes and communities, as opposed to just one species. By allowing the monitoring of spatio-temporal changes in community and metacommunity size, composition and functioning, the framework thus allows inference at local, landscape and macro scales.

Conservation and management plans tend to target either a species or region of conservation importance (Zipkin et al. 2010). Focusing on a single species frequently results in ignoring other interacting organisms, consequently introducing biases which are often not accounted for. When the focus is on a region, it is more efficient to look at community-level distribution and abundance patterns in order to design an appropriate management plan.

A common method for monitoring programs at the habitat level involves estimating species richness and composition, both of which can vary across the landscape due to biotic and abiotic factors. Due to this reason, it is important to consider species-specific detection when community-level studies are implemented. A multi-species occupancy approach is likely to provide better species richness and community composition estimates by simultaneously accounting for imperfect detection as well as species rarity and elusiveness. Furthermore, there are likely to be differences in the responses to habitat changes based on whether the taxa are rare or common. This too has implications on the detection of the species. Statistical tools to understand community-level dynamics and species-specific distribution estimates are gaining traction. However, so far, despite the increasing popularity of community models, incorporating biotic interactions is not common due to several reasons, including but not limited to logistical and statistical challenges.

With technological and statistical advances, and a growth in citizen science initiatives, distribution studies are resulting in increasingly large datasets that need to be managed (Hampton et al. 2013, Hines et al. 2015, Sun et al. 2019). There has also been an associated increase in software to manage such projects (Krishnappa and Turner 2014, Bubnicki et al. 2016, Niedballa et al. 2016, Ramachandran and Devarajan 2018, Thomson et al. 2018, Young et al. 2018).

The importance of including detailed information on (meta)data and following best practices is not only of concern to researchers. Evaluating the robustness of specific studies is also relevant to conservation biologists and wildlife managers who apply these tools, write transparent reports to inform decision-making processes, and ensure that their study follows an acknowledged protocol. In addition, having detailed information on (meta)data and followed best practices enhances the reusability of data (Wilkinson et al. 2016). In light of policy reporting needs and the recently proposed essential biodiversity variables (Pereira et al. 2013, Kissling et al. 2018), MSOMs can simultaneously provide estimates of community composition, species populations (i.e. occupancy), and species traits. These are essential measurement classes required for the study, reporting and management of biodiversity changes.

The linking of multi-faceted biodiversity metrics (i.e. taxonomic, phylogenetic, and functional) with spatial conservation science can help identify more efficient strategies for the conservation of biological diversity (Pollock et al. 2017). Given this, the proper use of analytical tools like MSOMs, that provide estimates of biodiversity metrics while accounting for imperfect detection, will be fundamental to robust inference.

Data availability statement

The dataset associated with the literature review has been deposited in the online data repository Figshare: <https://doi.org/10.6084/m9.figshare.11608449.v1> (Devarajan 2020).

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References

Beaudrot, L. et al. 2016. Standardized assessment of biodiversity trends in tropical forest protected areas: the end is not in sight. – PLoS Biol. 14: e1002357.

Broms, K. M. et al. 2016. Model selection and assessment for multispecies occupancy models. – Ecology 97: 1759–1770.

Bubnicki, J. W. et al. 2016. TRAPPER: an open source web-based application to manage camera trapping projects. – Methods Ecol. Evol. 7: 1209–1216.

Chambert, T. et al. 2015. Testing hypotheses on distribution shifts and changes in phenology of imperfectly detectable species. – Methods Ecol. Evol. 6: 638–647.

Devarajan, K. 2020. Multi-species occupancy models literature review dataset. – Figshare Digital Repository, <https://doi.org/10.6084/m9.figshare.11608449.v1>.

Dorazio, R. M. 2007. On the choice of statistical models for estimating occurrence and extinction from animal surveys. – Ecology 88: 2773–2782.

Dorazio, R. M. and Royle, J. A. 2005. Estimating size and composition of biological communities by modeling the occurrence of species. – J. Am. Stat. Assoc. 100: 389–398.

Dorazio, R. M. et al. 2006. Estimating species richness and accumulation by modeling species occurrence and detectability. – Ecology 87: 842–854.

Dorazio, R. M. et al. 2011. Modern methods of estimating biodiversity from presence–absence surveys. – In: Venora, G. et al. (eds), Biodiversity loss in a changing planet. InTech, pp. 277–302.

Geijzendorffer, I. R. et al. 2016. Bridging the gap between biodiversity data and policy reporting needs: an essential biodiversity variables perspective. – J. Appl. Ecol. 53: 1341–1350.

Genet, K. S. and Sargent, L. G. 2003. Evaluation of methods and data quality from a volunteer-based amphibian call survey. – Wildl. Soc. Bull. 31: 703–714.

Guillera-Arroita, G. 2017. Modelling of species distributions, range dynamics and communities under imperfect detection: advances, challenges and opportunities. – Ecography 40: 281–295.

Guillera-Arroita, G. et al. 2010. Design of occupancy studies with imperfect detection. – Methods Ecol. Evol. 1: 131–139.

Guillera-Arroita, G. et al. 2019. Inferring species richness using multi-species occupancy modeling: estimation performance and interpretation. – Ecol. Evol. 9: 780–792.

Hampton, S. C. et al. 2013. Big data and the future of ecology. – Front. Ecol. Environ. 11: 156–162.

Hines, G. et al. 2015. Aggregating user input in ecology citizen science projects. – Innov. Appl. Artific. Intell., pp. 3975–3980.

Iknayan, K. J. et al. 2014. Detecting diversity: emerging methods to estimate species diversity. – Trends Ecol. Evol. 29: 97–106.

Jarzyna, M. A. and Jetz, W. 2016. Detecting the multiple facets of biodiversity. – Trends Ecol. Evol. 31: 527–538.

Karenyi, N. et al. 2016. Incorporating species detectability into conservation targets based on species area relationship. – Divers. Distrib. 22: 758–769.

Kendall, W. L. et al. 2013. Relaxing the closure assumption in occupancy models: staggered arrival and departure times. – Ecology 94: 610–617.

Kéry, M. and Schaub, M. 2012. Bayesian population analysis using WinBUGS. – Elsevier.

Kéry, M. and Royle, J. A. 2015. Applied hierarchical modeling in ecology: analysis of distribution, abundance and species richness in R and BUGS: volume 1: prelude and static models. – Academic Press.

Kissling, W. D. et al. 2018. Towards global data products of essential biodiversity variables on species traits. – Nat. Ecol. Evol. 2: 1531–1540.

Kleijnen, J. P. 1998. Experimental design for sensitivity analysis, optimization and validation of simulation models. – In: Banks, J. (ed.), Handbook of simulation. Wiley, pp. 173–223.

Krishnappa, Y. S. and Turner, W. C. 2014. Software for minimalistic data management in large camera trap studies. – Ecol. Inform. 24: 11–16.

Linden, D. W. et al. 2017. Examining the occupancy–density relationship for a low-density carnivore. – J. Appl. Ecol. 54: 2043–2052.

MacKenzie, D. I. et al. 2002. Estimating site occupancy rates when detection probabilities are less than one. – Ecology 83: 2248–2255.

MacKenzie, D. I. et al. 2004. Investigating species co-occurrence patterns when species are detected imperfectly. – J. Anim. Ecol. 73: 546–555.

MacKenzie, D. I. et al. 2017. Occupancy estimation and modeling: inferring patterns and dynamics of species occurrence. – Elsevier.

McClintock, B. T. et al. 2010. Unmodeled observation error induces bias when inferring patterns and dynamics of species occurrence via aural detections. – Ecology 91: 2446–2454.

McNew, L. B. and Handel, C. M. 2015. Evaluating species richness: biased ecological inference results from spatial heterogeneity in detection probabilities. – Ecol. Appl. 25: 1669–1680.

Niedballa, J. et al. 2016. camtrapR: an R package for efficient camera trap data management. – Methods Ecol. Evol. 7: 1457–1462.

Ovaskainen, O. et al. 2017. How to make more out of community data? A conceptual framework and its implementation as models and software. – Ecol. Lett. 20: 561–576.

Peck, S. L. 2004. Simulation as experiment: a philosophical reassessment for biological modeling. – Trends Ecol. Evol. 19: 530–534.

Pereira, H. M. et al. 2013. Essential biodiversity variables. – Science 339: 277–278.

Pollock, K. H. 1982. A capture–recapture design robust to unequal probability of capture. – J. Wildl. Manage. 46: 752–757.

Pollock, L. J. et al. 2014. Understanding co-occurrence by modelling species simultaneously with a joint species distribution model (JSDM). – Methods Ecol. Evol. 5: 397–406.

Pollock, L. J. et al. 2017. Large conservation gains possible for global biodiversity facets. – Nature 546: 141.

Powers, S. M. and Hampton, S. E. 2019. Open science, reproducibility and transparency in ecology. – Ecol. Appl. 29: e01822.

Ramachandran, P. and Devarajan, K. 2018. Vixen: an open-source package for managing multimedia data. – Methods Ecol. Evol. 9: 785–792.

Rota, C. T. et al. 2009. Occupancy estimation and the closure assumption. – J. Appl. Ecol. 46: 1173–1181.

Rota, C. T. et al. 2016. A multispecies occupancy model for two or more interacting species. – Methods Ecol. Evol. 7: 1164–1173.

Royle, J. A. and Link, W. A. 2006. Generalized site occupancy models allowing for false positive and false negative errors. – Ecology 87: 835–841.

RStudio Team 2018. RStudio: integrated development for R. – RStudio, <www.rstudio.com/>.

16000587, 2020, 11, Downloaded from https://nsojournals.onlinelibrary.wiley.com/doi/10.1111/ecog.04881 by Cochrane Chile, Wiley Online Library on [12/04/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Ecography E4 award

Simons, T. R. et al. 2007. Experimental analysis of the auditory detection process on avian point counts. – Auk 124: 986–999.
Steenweg, R. et al. 2019. Species-specific differences in detection and occupancy probabilities help drive ability to detect trends in occupancy. – Ecosphere 10: e02639.
Sun, C. C. et al. 2019. Incorporating citizen science data in spatially explicit integrated population models. – Ecology 100: e02777.
Tenan, S. et al. 2014. Bayesian model selection: the steepest mountain to climb. – Ecol. Model. 283: 62–69.
Thomson, R. et al. 2018. Closing the gap between camera trap software development and the user community. – Afr. J. Ecol. 56: 721–739.
Tobler, M. W. et al. 2015. Spatiotemporal hierarchical modelling of species richness and occupancy using camera trap data. – J. Appl. Ecol. 52: 413–421.
Tobler, M. W. et al. 2019. Joint species distribution models with species correlations and imperfect detection. – Ecology 100: e02754.
Waddle, J. H. et al. 2010. A new parameterization for estimating co-occurrence of interacting species. – Ecol. Appl. 20: 1467–1475.

Supplementary material (available online as Appendix ecog-04957 at <www.ecography.org/appendix/ecog-04957>). Appendix 1.

Wickham, H. 2010. ggplot2: elegant graphics for data analysis. – J. Stat. Softw. 35: 65–88.
Wickham, H. and Chang, W. 2008. ggplot2: an implementation of the grammar of graphics. – R package ver. 07, <https://CRAN.R-project.org/package=ggplot2>.
Wilkinson, M. D. et al. 2016. The FAIR guiding principles for scientific data management and stewardship. – Sci. Data 3: 160018.
Wright, W. J. et al. 2016. A goodness-of-fit test for occupancy models with correlated within-season revisits. – Ecol. Evol. 6: 5404–5415.
Young, S. et al. 2018. Software to facilitate and streamline camera trap data management: a review. – Ecol. Evol. 8: 9947–9957.
Zipkin, E. F. et al. 2009. Impacts of forest fragmentation on species richness: a hierarchical approach to community modelling. – J. Appl. Ecol. 46: 815–822.
Zipkin, E. F. et al. 2010. Multi-species occurrence models to evaluate the effects of conservation and management actions. – Biol. Conserv. 143: 479–484.

16000587, 2020, 11, Downloaded from https://nsojournals.onlinelibrary.wiley.com/doi/10.1111/ecog.04957, Wiley Online Library on [12/04/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License